

- 6 The cell membrane controls which substances can move into the cell or out of the cell.

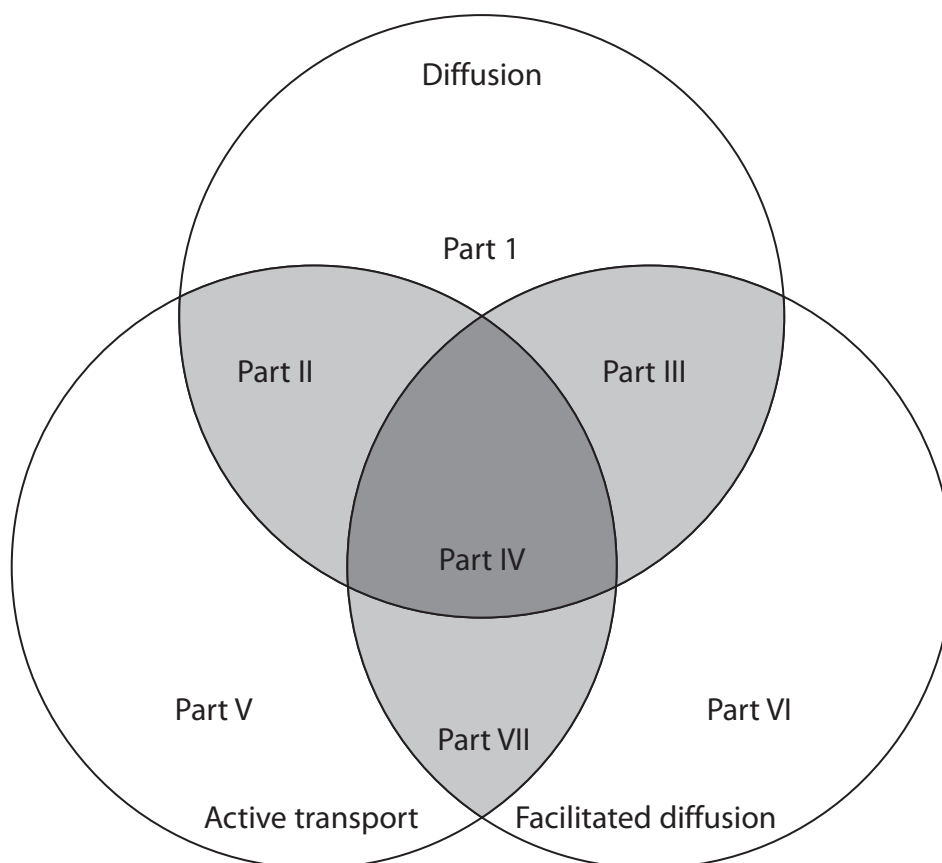
Processes by which substances can move into a cell or out of a cell include diffusion, facilitated diffusion, active transport and osmosis.

- (a) A Venn diagram can be used to show the similarities and differences between diffusion, facilitated diffusion and active transport.

Statements about similarities can be written in the numbered parts of the circles that overlap.

For example, statements about similarities shared by all three processes would be written in Part IV.

Statements about differences can be written in the numbered parts of the circles that do not overlap.



- (i) Which part would contain the statement: uses proteins?

(1)

- ☐ **A** II
- ☐ **B** III
- ☐ **C** IV
- ☐ **D** VII



(ii) Which part would contain the statement: needs energy in the form of ATP?

(1)

- ☐ **A** II
- ☐ **B** V
- ☐ **C** VI
- ☐ **D** VII

(iii) Which part would contain the statement: solutes can only move down a concentration gradient?

(1)

- ☐ **A** II
- ☐ **B** III
- ☐ **C** V
- ☐ **D** VII

(b) Osmosis can be defined as the movement of free water molecules through a partially permeable membrane, down a water potential gradient.

Explain this definition.

(3)

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*(c) The photograph shows a Chinese mitten crab.



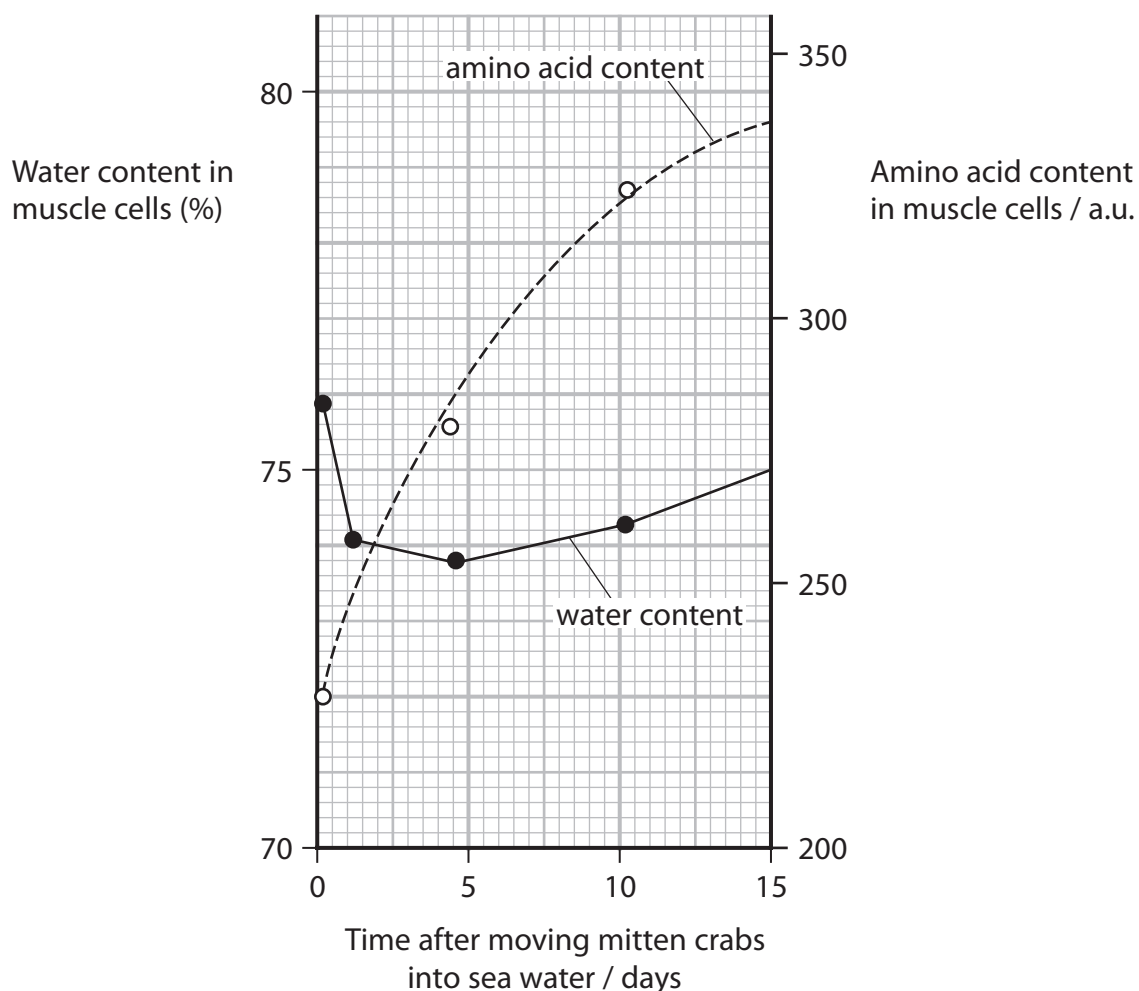
(Source: © WILDLIFE GmbH / Alamy Stock Photo)

Mitten crabs spend most of their lives in fresh water and only return to the sea to breed.

In an investigation, mitten crabs were kept in fresh water and then moved into sea water.

The water content and amino acid content in the muscle cells of these crabs were measured for 15 days after moving them from fresh water into sea water.

The graph shows the results of this investigation.



Sea water has a higher concentration of salt than fresh water.

Explain the changes in water content and amino acid content in the muscle cells of the crabs in this investigation.

(6)

(Total for Question 6 = 12 marks)

